fn sample_bernoulli_exp

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This proof resides in **"contrib"** because it has not completed the vetting process.

Proves soundness of `fn sample_bernoulli_exp` in `mod.rs` at commit 0be3ab3e6 (up to date). This proof is an adaptation of subsection 5.1 of [CKS20].

1 Vetting history

- Pull Request #519

2 Pseudocode

Precondition

$\mathtt{x} \in \mathbb{Q} \wedge \mathtt{x} > 0$

Implementation

```
def sample_bernoulli_exp(x) -> bool:
    while x >= 1:
        if sample_bernoulli_exp1(1):
            x -= 1
        else:
            return False
    return sample_bernoulli_exp1(x)
```

Postcondition

`sample_bernoulli_exp` returns a boolean sample from $Bernoulli(exp(-x))$, or an error propagated from `sample_bernoulli_exp1`.

3 Proof

Let r be the return value of `sample_bernoulli_exp(x)`, assuming the preconditions are met, and the function does not return an error.

Theorem 3.1. r is a realization of $R \sim Bernoulli(exp(-x))$, where $x \in \mathbb{Q}$ and $x > 0$ [CKS20].

Proof. Let $B_i \sim Bernoulli(exp(-1))$ be the sequence of booleans returned from `sample_bernoulli_exp1(1)`, and $C \sim Bernoulli(exp(-(x - \lfloor x \rfloor)))$. The function only returns $\top$ if $B_i = \top$ for all i and $C = \top$.

$$
\begin{aligned}
P[R = \top] &= P[B_1 = B_2 = ... = B_{\lfloor x \rfloor} = C = \top] \\
&= \prod_{i=1}^{\lfloor x \rfloor} P[B_i = \top] P[C = \top] && \text{All } B_i \text{ are independent.} \\
&= exp(-1)^{\lfloor x \rfloor} exp(\lfloor x \rfloor - x) \\
&= exp(-x)
\end{aligned}
$$

□

References

[CKS20] Clément L. Canonne, Gautam Kamath, and Thomas Steinke. The discrete gaussian for differential privacy. *CoRR*, abs/2004.00010, 2020.